

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
 mendamar.badral@gmail.com |  +36 70 7385 744 |  Budapest, Hungary

Driven by a problem solving mindset and fueled by curiosity, I always aspire to develop genuine solutions for the real-world, impactful problems.

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering
Supervisor: [Márton Vaitkus](#)
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics
GPA: 3.2/4.0 (rank 3/34)
Supervisor: [Galtbayar Artbazar](#)
Thesis: [Evaluating land degradation using image processing methods](#)

EXPERIENCE

Center of Mathematics Application Research Lab Oct, 2023 – June, 2025
Undergraduate Lab Lead & Research Intern

- **Technical Leadership:** Selected as Lab Lead for an 8-student research team; directed weekly seminars to break down complex machine learning and statistical modeling concepts.
- **Spatial Analysis & Modeling:** Developed a custom Python application to process sensor data and perform statistical diagnostics, utilizing Moran's I coefficient to detect and model spatial clustering in high-dimensional datasets.
- **Simulation:** Modeled logistics workflows in a [cargo delivery simulation](#), identifying operational inefficiencies and projecting a 23% reduction in overhead costs.

PROJECTS

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Implemented polynomial regression models from Bishop & Bishop's *Deep Learning Foundations and Concepts* to study **overfitting, underfitting, and the bias-variance tradeoff**.
- Generated synthetic datasets and fit models of varying polynomial degree to analyze how model complexity affects generalization error.

[Short Animation: What is a Derivative?](#)

- Created a short-form educational video explaining the mathematical concept of derivatives utilizing Python [Manim](#) library. Achieved 100 views on the YouTube platform.

AWARDS & SCHOLARSHIPS

Stipendium Hungaricum Scholarship Sep, 2025 – Present

SKILLS

Languages: English (C1 [IELTS 8.0](#)), Mongolian (native), Hungarian (A2)
Programming: Python, Bash, Julia, C/C++, R, MATLAB
Tools: Git, GitHub, Linux, Docker, Manim, $\text{\LaTeX}$